5 **(a)** Ernest Rutherford proposed a planetary model for the hydrogen atom. In the model, a single electron is treated as a point-like charged particle, moving in circular motion around a stationary proton (the nucleus) as shown in Fig. 5.1.

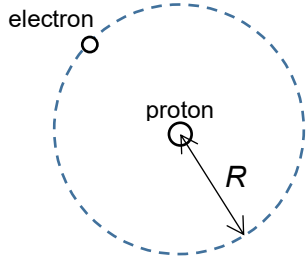

Fig. 5.1 (not to scale)

(i) State two forms of energy that the system possesses and state whether each form is by convention, positive or negative.

.....

..... [2]

(ii) Explain why the total energy of the system is conventionally taken to be negative when the electron is orbiting the nucleus.

.....

.....

.....

..... [2]

(b) In quantum mechanics, it is not possible to model the path of the electron in the hydrogen atom to be a well-defined circle around the proton.

(i) Using Heisenberg uncertainty principle for position and momentum, explain why the path of the electron inside the atom is not well-defined.

.....

.....

.....

..... [2]

(ii) Despite not knowing the path of the electron, we are certain that the electron remains bound to the nucleus. By considering your answer to **(a)**, show that the maximum kinetic energy of the bound electron is given by $E_k = \dfrac{e^2}{4\pi\varepsilon_0 R}$. Explain your working clearly. [2]